MediSync AI: Intelligent Hospital Resource and Patient Flow Management System

Industry ContextSCENARIO BASED REQUIREMENT ANALYSIS REPORT

Submitted in the partial fulfilment for the Course of

CSA1012- Software Engineering

to the award of the degree of

BACHELOR OF ENGINEERING

IN

Artificial Intelligence and Machine Learning

Submitted by

Maniyam Anushka (192525411)
Under the Supervision of

Dr C Anita Davamani

SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105

1. Introduction

1.1 Purpose

This Software Requirement Specification (SRS) document defines the functional and non-functional requirements for MediSync AI, an AI-powered hospital resource and patient flow management system. It is intended to guide the design, development, and validation of the system by clearly documenting stakeholder needs, system behavior, actors, use cases, and constraints, in accordance with the scenario-based requirement analysis assigned for CO2.

1.2 Business Domain

The system operates in the healthcare and hospital administration domain, specifically addressing hospital operations management — bed management, staff scheduling, medical equipment tracking, patient admission/discharge workflows, and interdepartmental patient transfers. It combines Artificial Intelligence, Internet of Things (IoT), and predictive analytics to support smart, data-driven hospital operations.

1.3 Stakeholders

- Hospital Administrators — responsible for overall resource planning, capacity management, and operational decision-making.
- Doctors / Physicians — responsible for patient treatment decisions, transfer approvals, and discharge planning.
- Nurses / Ward Staff — responsible for day-to-day bed, equipment, and patient monitoring at the ward level.
- Patients — recipients of care who interact with the system for appointments and status visibility.
- IT / Hospital Information Systems Team — responsible for system integration, maintenance, and data governance.
- Regulatory Bodies — set compliance requirements for healthcare data privacy and security (e.g., HIPAA-equivalent standards).
- Hospital Management / Board — funds the initiative and evaluates return on investment through operational efficiency gains.

1.4 Existing Challenges

Hospitals currently face overcrowding, prolonged patient waiting times, inefficient bed utilization, and uneven distribution of medical resources. Existing hospital information systems mainly manage administrative records and lack intelligent decision-support capabilities for optimizing patient flow. Delays in transfers, diagnostics, and discharge planning reduce effective hospital capacity and negatively affect care quality.

1.5 System Objectives

1. Develop an AI-powered platform for intelligent hospital resource and patient flow management.
2. Monitor real-time availability of beds, medical equipment, and healthcare personnel.
3. Predict patient admissions, discharge times, and resource demand using machine learning.
4. Optimize patient scheduling, admissions, and interdepartmental transfers.
5. Reduce patient waiting times through intelligent workflow management.
6. Provide real-time dashboards for hospital administrators and medical staff.
7. Generate predictive analytics to support emergency preparedness and capacity planning.
8. Improve healthcare efficiency while maintaining high-quality patient care.

2. Problem Description

Hospitals must effectively manage beds, medical equipment, healthcare professionals, and patient appointments to ensure quality care and reduce operational costs. Traditional hospital management systems function independently across departments, making it difficult to coordinate patient movement between the emergency department, outpatient clinics, laboratories, and inpatient wards. This lack of coordination results in overcrowding, delayed treatment, inefficient use of scarce resources, and reduced patient satisfaction. There is a need for an AI-powered hospital management platform that continuously analyzes patient movement, predicts resource requirements, optimizes scheduling, and recommends efficient allocation of beds, staff, and equipment.

3. Scope of the System

MediSync AI will provide an integrated, AI-driven platform covering:

- Real-time monitoring of bed, staff, and equipment availability across all hospital departments.

- Predictive analytics for patient admission volume, discharge timing, and resource demand.
- Decision-support recommendations for bed allocation, staff assignment, and patient transfers.
- Role-based dashboards for administrators, doctors, and nurses.
- Patient-facing appointment booking and status visibility.
- Integration with existing hospital systems (Laboratory Information System, Billing System) via standard interfaces.

Out of scope: the system does not replace core Electronic Health Record (EHR) clinical documentation, does not perform direct medical diagnosis, and does not process financial transactions beyond utilization data hand-off to the billing system.

4. Functional Requirements

The following functional requirements were derived from the problem statement and system objectives.

ID	Requirement Description
FR-1	The system shall allow authorized users (Administrator, Doctor, Nurse) to log in and be authenticated based on role-based access control.
FR-2	The system shall capture and display real-time availability of hospital beds (ICU, general ward, emergency, isolation) across all departments.
FR-3	The system shall track real-time availability and location of medical equipment (ventilators, monitors, wheelchairs, diagnostic machines) using IoT sensor data.
FR-4	The system shall track real-time availability of healthcare personnel (doctors, nurses, technicians) by department and shift.
FR-5	The system shall use machine learning models to predict patient admission volume, discharge timing, and resource demand for defined future time windows.
FR-6	The system shall recommend optimal bed allocation and interdepartmental patient transfers based on predicted demand and current occupancy.
FR-7	The system shall allow doctors and administrators to approve, modify, or reject system-recommended transfers and allocations.

ID	Requirement Description
FR-8	The system shall allow patients to book, view, and reschedule outpatient appointments.
FR-9	The system shall generate discharge planning recommendations based on treatment progress and predicted bed demand.
FR-10	The system shall provide real-time interactive dashboards showing bed occupancy, staff allocation, equipment status, and patient flow metrics to administrators and medical staff.
FR-11	The system shall generate predictive analytics reports (daily, weekly, and emergency-surge forecasts) for capacity planning.
FR-12	The system shall send automated alerts and notifications to relevant staff for bed shortages, equipment unavailability, or predicted surges.
FR-13	The system shall interface with the hospital Laboratory Information System to retrieve diagnostic and test results relevant to patient flow decisions.
FR-14	The system shall interface with the Billing System to update resource-utilization records used for cost calculation.
FR-15	The system shall maintain an audit log of all resource-allocation decisions, overrides, and system recommendations.

5. Non-Functional Requirements

Category	Requirement Description
Performance	The system shall refresh bed, staff, and equipment availability data within 5 seconds of a status change and generate predictive forecasts within 10 seconds of a request.
Scalability	The system shall support concurrent use by at least 500 hospital staff users and scale to accommodate multi-hospital / multi-campus deployments without redesign.
Reliability	The system shall maintain 99.5% uptime for core patient-flow and dashboard modules, with automatic failover for critical monitoring services.
Security	The system shall encrypt all patient and operational data in transit and at rest, enforce role-based access control, and comply with applicable healthcare data-protection regulations (e.g., HIPAA-equivalent standards).
Usability	The system shall provide an intuitive, role-specific interface requiring no more than 2 hours of training for clinical staff to perform core tasks.
Maintainability	The system architecture shall be modular (microservices-based) to allow independent updates to the prediction engine, dashboard, and interface layers.
Availability	Critical modules (bed monitoring, alerts) shall be available 24/7, including during scheduled maintenance of non-critical modules.
Interoperability	The system shall support standard healthcare data-exchange formats (e.g., HL7/FHIR) for integration with existing hospital information systems.
Data Accuracy	Predictive models shall be validated to maintain a minimum forecast accuracy threshold (e.g., ±10% for admission volume) and be retrained periodically.

Category	Requirement Description
Auditability	All allocation decisions and manual overrides shall be logged with timestamp and user ID for compliance review.

6. Actors and Use Cases

6.1 Primary Actors

- Hospital Administrator — configures resource pools, reviews dashboards, approves optimization recommendations, generates reports.
- Doctor / Physician — reviews patient status, approves transfers/discharges, views dashboards.
- Nurse / Ward Staff — updates bed/equipment status, receives alerts, executes transfer instructions.
- Patient — books/reschedules appointments, views own queue/status.

6.2 Secondary (Supporting) Actors

- AI/ML Prediction Engine — internal service generating admission, discharge, and demand forecasts.
- IoT Sensor Network — provides real-time bed and equipment status data.
- Laboratory Information System (external) — supplies diagnostic results relevant to flow decisions.
- Billing System (external) — receives resource-utilization data for cost processing.

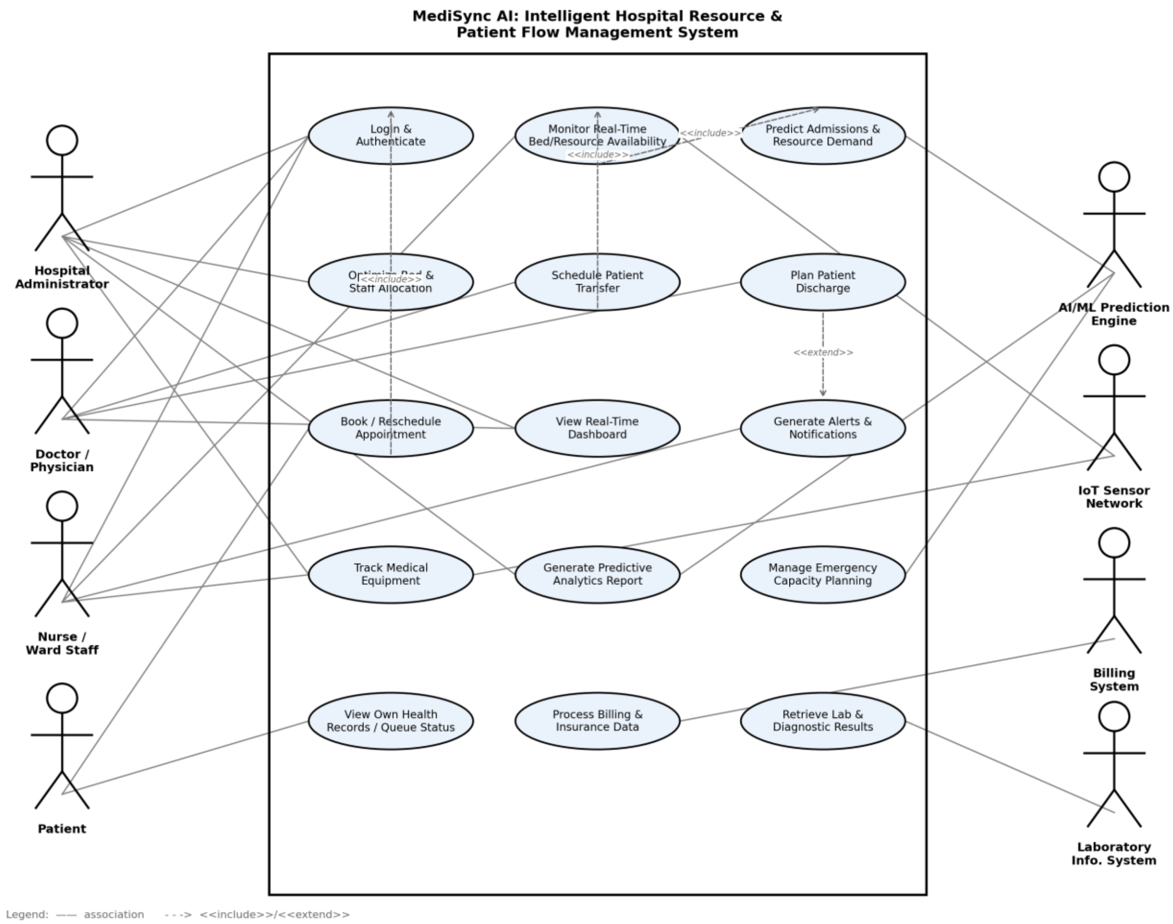
6.3 Use Cases by Actor

Actor	Associated Use Cases
Hospital Administrator	Login & Authenticate; Optimize Bed & Staff Allocation; View Real-Time Dashboard; Generate Predictive Analytics Report; Track Medical Equipment
Doctor / Physician	Login & Authenticate; Schedule Patient Transfer; Plan Patient Discharge; View Real-Time Dashboard
Nurse / Ward Staff	Login & Authenticate; Monitor Real-Time Bed/Resource Availability; Generate Alerts & Notifications; Track Medical Equipment

Actor	Associated Use Cases
Patient	Book / Reschedule Appointment; View Own Health Records / Queue Status
AI/ML Prediction Engine	Predict Admissions & Resource Demand; Generate Predictive Analytics Report; Manage Emergency Capacity Planning
IoT Sensor Network	Monitor Real-Time Bed/Resource Availability; Track Medical Equipment
Billing System	Process Billing & Insurance Data
Laboratory Information System	Retrieve Lab & Diagnostic Results

6.4 Use Case Diagram

The diagram below represents the interactions between actors and MediSync AI, including include/extend relationships between key use cases.


MediSync AI: Intelligent Hospital Resource &
Patient Flow Management System
Hospital
Administrator
Doctor /
Physician
Nurse /
Ward Staff
Patient
AI/ML Prediction
Engine
IoT Sensor
Network
Billing
System
Laboratory
Info. System
Login &
Authenticate
Monitor Real-Time
Bed/Resource Availability
Predict Admissions &
Resource Demand
Optimize Bed &
Staff Allocation
Schedule Patient
Transfer
Plan Patient
Discharge
Book / Reschedule
Appointment
View Real-Time
Dashboard
Generate Alerts &
Notifications
Track Medical
Equipment
Generate Predictive
Analytics Report
Manage Emergency
Capacity Planning
View Own Health
Records / Queue Status
Process Billing &
Insurance Data
Retrieve Lab &
Diagnostic Results
<<include>>
<<include>>
<<include>>
<<extend>>
Legend: —— association - - -> <<include>>/<<extend>>

6.5 Detailed Use Case Descriptions

Representative use cases are detailed below; remaining use cases follow the same structure.

UC-01: Predict Admissions & Resource Demand

Use Case ID	UC-01
Use Case Name	Predict Admissions & Resource Demand
Actors	AI/ML Prediction Engine, Hospital Administrator
Preconditions	Historical and real-time patient-flow data is available to the prediction engine.
Main Flow	1. The AI/ML engine ingests real-time bed, staff, and admission data. 2. The engine runs forecasting models for a defined time horizon. 3. The system generates predicted admission volume, discharge timing, and resource demand. 4. Results are pushed to the dashboard and used by the allocation-optimization use case.
Postconditions	Updated forecast is available on the administrator dashboard and stored for reporting.

UC-02: Optimize Bed & Staff Allocation

Use Case ID	UC-02
Use Case Name	Optimize Bed & Staff Allocation
Actors	Hospital Administrator, AI/ML Prediction Engine
Preconditions	Predicted demand (UC-01) and current occupancy data are available.
Main Flow	1. Administrator opens the allocation-optimization view. 2. System compares predicted demand against current bed/staff availability. 3. System recommends allocation/reallocation actions. 4. Administrator approves, modifies, or rejects recommendations. 5. Approved changes are applied and staff are notified.

Postconditions	Bed and staff assignments are updated; notifications are sent to affected wards.

UC-03: Schedule Patient Transfer

Use Case ID	UC-03
Use Case Name	Schedule Patient Transfer
Actors	Doctor / Physician, Nurse / Ward Staff
Preconditions	Patient is admitted and requires transfer between departments; bed availability data is current (includes UC — Monitor Bed Availability).
Main Flow	1. Doctor initiates a transfer request for a patient. 2. System checks destination-ward bed and staff availability. 3. System confirms or suggests an alternative destination/time. 4. Doctor confirms the transfer. 5. Ward staff receive transfer instructions and update bed status.
Postconditions	Patient transfer is scheduled and reflected in real-time bed availability.

UC-04: Book / Reschedule Appointment

Use Case ID	UC-04
Use Case Name	Book / Reschedule Appointment
Actors	Patient
Preconditions	Patient is registered and authenticated (includes UC — Login & Authenticate).
Main Flow	1. Patient logs into the patient portal. 2. Patient selects department, doctor, and preferred time slot. 3. System checks slot availability and confirms or offers alternatives. 4. Patient confirms booking or reschedules an existing appointment.

Postconditions	Appointment is created/updated and reflected in the department schedule.

7. Data Requirements

The system shall create, store, update, and retrieve the following categories of data:

- Patient data — patient ID, admission/discharge timestamps, department, treatment status (linked to, but not duplicating, core EHR records).
- Bed and ward data — bed ID, ward, status (occupied/vacant/reserved/cleaning), bed type (ICU, general, isolation).
- Staff data — staff ID, role, department, shift timing, current assignment.
- Equipment data — equipment ID, type, location, availability status, maintenance schedule.
- Predictive model data — historical admission trends, forecast outputs, model accuracy metrics.
- Appointment data — patient ID, department, doctor, scheduled date/time, status.
- Alert/notification logs — alert type, timestamp, recipient, acknowledgement status.
- Audit data — user ID, action performed, timestamp, before/after values for allocation decisions.

8. External Interface Requirements

8.1 User Interfaces

- Web-based administrator and clinical dashboard (desktop-optimized, responsive).
- Mobile-friendly interface for nurses and doctors for on-the-go alerts and approvals.
- Patient-facing web/mobile portal for appointment booking.

8.2 Hardware Interfaces

- IoT sensors and RFID/BLE tags on beds and equipment for real-time location and status tracking.

8.3 Software Interfaces

- Integration with the hospital's Laboratory Information System via HL7/FHIR-compliant APIs.
- Integration with the Billing System for resource-utilization data exchange.
- Integration with existing Hospital Information System (HIS) for patient admission records.

8.4 Communication Interfaces

- Secure REST/HTTPS APIs for system-to-system communication.
- Push notification and SMS/email gateway for staff alerts.

9. System Features

Feature	Description
Real-Time Resource Monitoring	Live view of bed, staff, and equipment status sourced from IoT sensors and manual updates.
Predictive Analytics Engine	ML-based forecasting of admissions, discharges, and resource demand.
Intelligent Allocation Recommendations	System-generated, human-approvable recommendations for bed/staff optimization and transfers.
Role-Based Dashboards	Customized dashboard views for administrators, doctors, and nurses.
Alerts & Notifications	Automated alerts for shortages, predicted surges, and pending approvals.
Patient Self-Service Portal	Appointment booking, rescheduling, and status visibility for patients.
Reporting Module	Scheduled and on-demand predictive and operational reports for capacity planning.
Audit & Compliance Logging	Full traceability of allocation decisions and manual overrides.

10. Assumptions and Constraints

10.1 Assumptions

- Hospital departments have reliable network connectivity to support real-time data exchange.
- IoT sensors/tags will be installed and maintained on beds and critical equipment.
- Sufficient historical patient-flow data is available to train reliable predictive models.

- Hospital staff will receive basic training and are willing to adopt AI-assisted recommendations.
- Existing hospital systems (HIS, LIS, Billing) expose integrable APIs or data-exchange interfaces.
- Final allocation decisions remain subject to human (clinical) approval; the system is advisory, not autonomous, for clinical actions.

10.2 Constraints

Type	Constraint
Technical	Must integrate with legacy hospital information systems that may have limited or outdated APIs.
Operational	System must operate continuously (24/7) without disrupting existing emergency and ward workflows during rollout.
Economic	Implementation budget limits the scope of IoT hardware deployment in the initial phase.
Legal / Regulatory	Must comply with healthcare data-privacy regulations (e.g., HIPAA-equivalent) for storage and transmission of patient data.
Time	Initial deployment (pilot phase) must be completed within the academic/project timeline defined for the assignment scenario.
Security	All system access must be role-based and auditable; no direct patient data exposure to unauthorized external systems.

11. Requirement Completeness and Consistency Validation

11.1 Stakeholder Coverage Check

Each identified stakeholder (Administrator, Doctor, Nurse, Patient, IT team, Regulators, Management) is mapped to at least one functional requirement or use case, confirming that no primary stakeholder need has been omitted from FR-1 through FR-15 and the associated use cases.

11.2 Ambiguity and Redundancy Check

- Ambiguity: Requirement FR-5 originally stated a vague forecast horizon; it has been clarified in the NFR table (Data Accuracy) with a measurable accuracy threshold (±10%) to remove ambiguity.
- Redundancy: FR-2, FR-3, and FR-4 were reviewed to ensure bed, equipment, and staff monitoring are documented as distinct requirements rather than overlapping duplicates, since each tracks a different resource type.
- Consistency: Use cases (Section 6.3) were cross-checked against functional requirements (Section 4) to confirm each use case is traceable to at least one FR, and each FR is exercised by at least one use case.

11.3 Missing Requirement Check

- Added an audit-logging requirement (FR-15) after review, since accountability for AI-driven recommendations was identified as a gap during validation.
- Added interoperability (HL7/FHIR) as a non-functional requirement, since the initial draft assumed integration without specifying a standard.

11.4 Suggested Improvements

- Introduce a formal requirement-prioritization exercise (e.g., MoSCoW) before design to sequence the 15 functional requirements across release phases.
- Define explicit Service Level Agreements (SLAs) for third-party integrations (LIS, Billing) to avoid ambiguity in external interface expectations.
- Conduct stakeholder walkthroughs of the use case diagram to validate actor-use case mappings before proceeding to design.